Expert-Augmented Causal SHAP: recovering DAG-consistent feature importance via iterative causal discovery and domain knowledge.

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Abstract

Feature importance methods such as SHAP are widely used to help explain machine learning models in health research and other areas. Yet these methods can misattribute feature importance when features have causal structure. Specifically, mediators can absorb credit from upstream causes — a topological artifact that inflates their apparent importance at the expense of upstream causal drivers.

We propose an expert-augmented workflow for DAG-consistent feature attribution that combines constrained causal discovery with an interactive DAG interface to resolve ambiguous edges before computing causal SHAP. The workflow supports iterative refinement through required and forbidden edges and compares standard, DAG-constrained, and adjustment-set SHAP.

We evaluated the approach in synthetic data generated under known DAGs using the R package *simcausal* and in real-world MIMIC-IV data. In simulations, standard SHAP over-attributed importance to mediators, whereas causal SHAP re-ranked features (Kendall's $\tau=0.42$ between methods) and reduced mediator inflation. In MIMIC-IV, the expert-informed DAG helped distinguish upstream severity drivers from downstream interventions. These results support expert-guided DAG refinement as a practical route to more causally coherent explanations in structured clinical machine learning.

WHAT GOES WRONG

- Mediators absorb credit from their upstream causes
- Downstream consequences look like risk factors
- Conditioning on colliders opens spurious paths

THE FIX

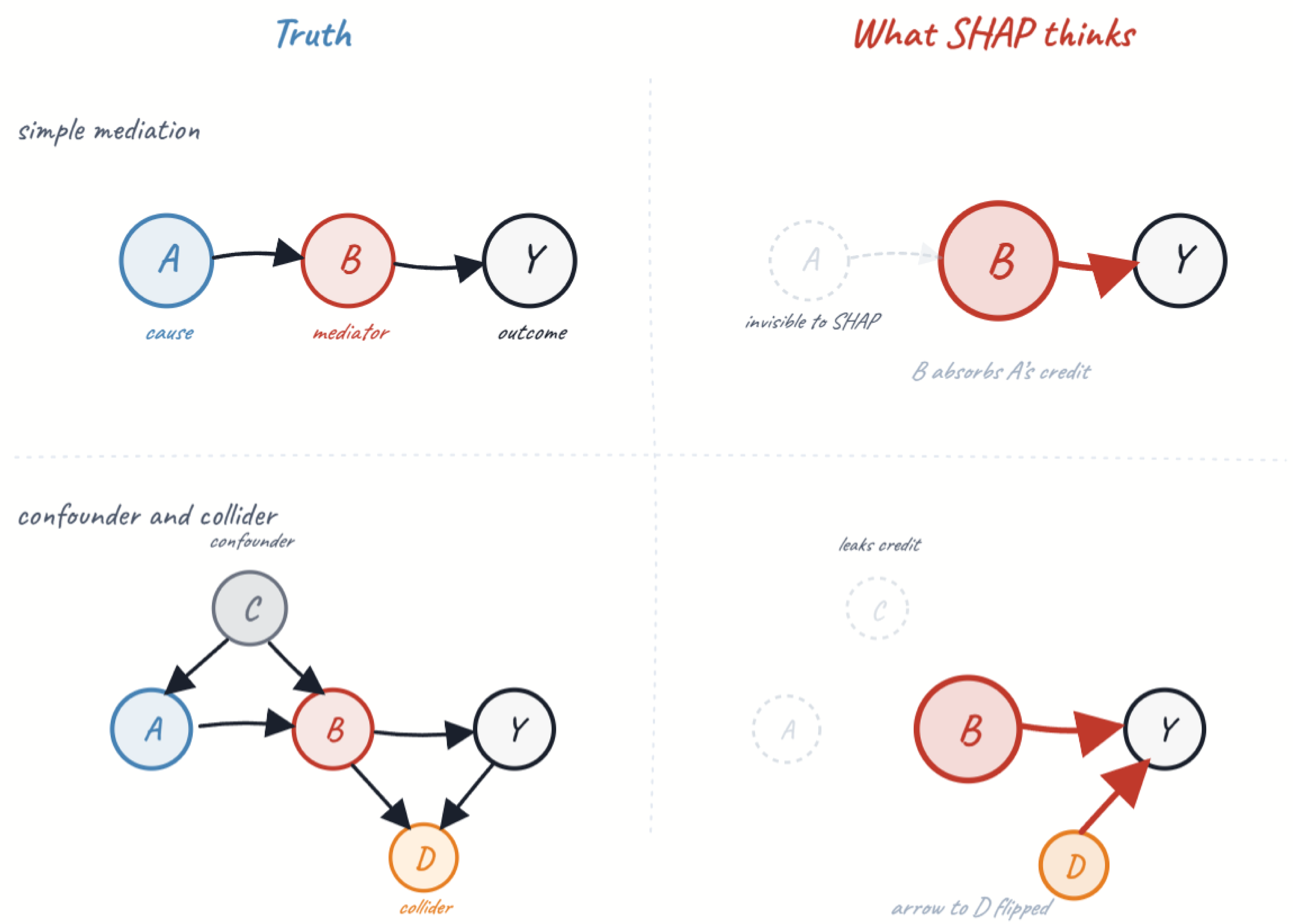
- Restrict SHAP to topological orderings of a DAG
- Replace observational with interventional expectations
- Resolve ambiguous edges with domain expertise

SHAP has short memory.

Downstream variables *HOG* credit!

THE CENTRAL ARGUMENT

Truth vs. what SHAP thinks



The closer a variable sits to Y, the more credit SHAP gives it — even when it isn't the cause.

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WHY IT LEAKS CREDIT

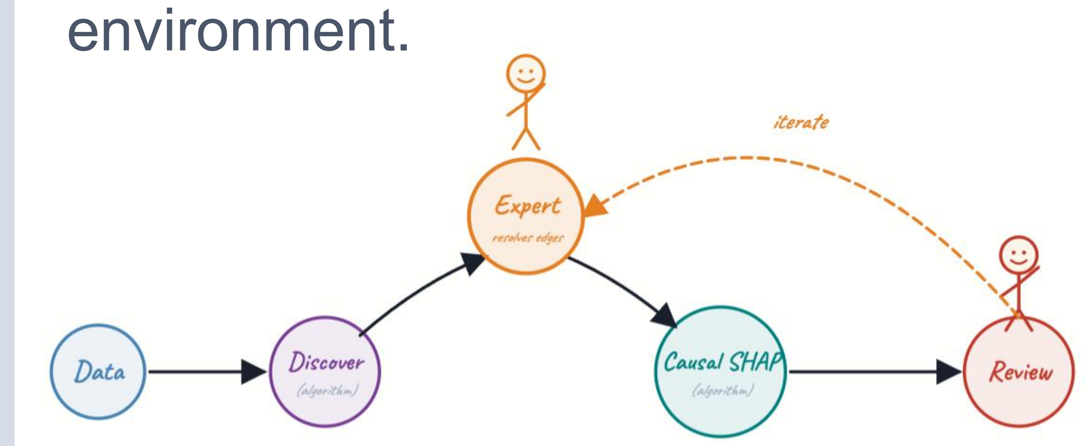
Every ordering gets an equal vote.

$$\varphi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n-|S|-1)!}{n!}\left[f(S \cup \{i\}) - f(S)\right]$$

The sum runs over *every* permutation, including orderings where a mediator enters before its upstream cause. When the two are tightly coupled, the mediator's marginal contribution looks large whenever it arrives first. Averaging bakes that leakage into the attribution by design.

Shapley 1953 · Lundberg & Lee, NeurIPS 2017

Causal SHAP is not an automated pipeline. It's a disciplined collaboration: the discovery algorithm proposes, the expert disposes, and the attribution is grounded in a DAG that both parties can defend. The app (Python Shiny) threads these five steps into a single interactive environment.



No existing tool formalizes this iterative expert-in-the-loop refinement from CPDAG to resolved DAG to causal attribution. The value isn't any single step → it's the cycle, with immediate visual feedback on how DAG edits change the attributions.

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